

CSE 321
Quiz 02
Set A
Time: 35 minutes

Given the arrival time and burst time for the processes below, draw a **gantt chart** using multilevel feedback queue scheduling algorithm showing the states of the ready queues of different levels. **Then calculate the average waiting time, response time and turnaround time.**

There are 3 queues with decreasing priority where the priority queue 0 and 1 is using round robin and the last queue is using the shortest job first. **Q1 (quantum: 5, priority:0), Q2 (quantum: 8, priority: 1), Q3 (SJF [preemptive], priority:2).**

In case of I/O request completion, a process will be promoted one level upon I/O completion.

Processes	Arrival Time	Burst Time	I/O Request
P1	0	10	3s IO after 7s CPU
P2	2	23	4s I/O after 5s CPU 3s I/O after 12s CPU
P3	5	18	N/A
P4	7	16	2s I/O after 11s CPU